Test of Hypothesis
Part-1
(Large Samples Tests)
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Hypothesis Testing: Steps

Use of the following sequence of steps in applying hypothesis testing.

1. **From the problem context, identify the parameter of interest.**
2. **State the null hypothesis, H_0.**
3. **Specify an appropriate alternative hypothesis, H_1.**
4. **Choose a significance level .**
5. **Determine an appropriate test statistic.**
6. **State the rejection region for the statistic.**
7. Compute any necessary sample quantities, substitute these into the equation for **the test statistic, and compute that value.**
8. **Decide whether or not H_0 *should be rejected and report that in the problem context.***

Follow like this:

Example: Test the assumption that the true mean life of TV sets in US homes is at least 3 years.

- **1. State H_0** $\quad H_0: \mu = 3$
- **2. State H_1** $\quad H_1: \mu < 3$
- **3. Choose α** $\quad \alpha = 0.05$
- **4 Set Up Critical Value(s)** $Z_{tab} = -1.645$
- **5 Compute Test Statistic** $Z_{Cal} = -2$
- **6 Make Statistical Decision:** $|Z_{cal}| > |Z_{tab}|$

Test statics falls in rejection region. So, *Reject Null H_0*

- **7. Express Decision:** ***The true mean of TV set is less than 3 in the US households.***

LoS & Critical Values of Z statistic

CRITICAL VALUES (z_α) OF Z

Critical Values (z_α)	Level of significance (α) 1%	5%	10%
Two-tailed test	$\lvert Z_\alpha \rvert = 2{\cdot}58$	$\lvert Z_\alpha \rvert = 1{\cdot}96$	$\lvert Z_\alpha \rvert = 1{\cdot}645$
Right-tailed test	$Z_\alpha = 2{\cdot}33$	$Z_\alpha = 1{\cdot}645$	$Z_\alpha = 1{\cdot}28$
Left-tailed test	$Z_\alpha = -2{\cdot}33$	$Z_\alpha = -1{\cdot}645$	$Z_\alpha = -1{\cdot}28$

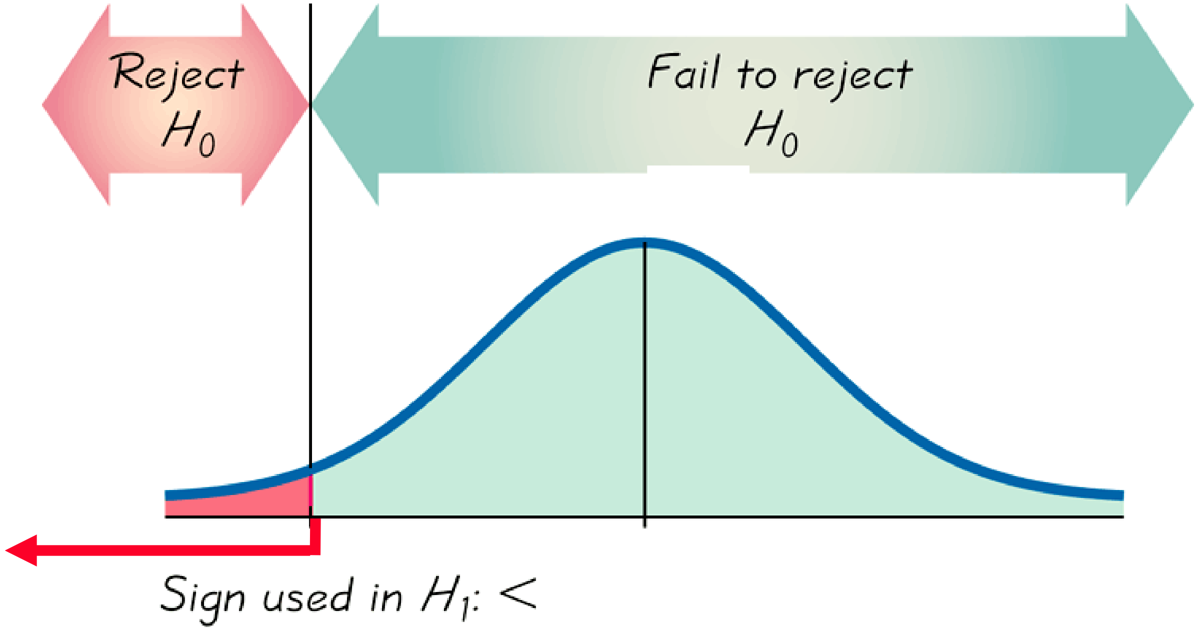
Left-tailed Test

H_0: =

H_1: <

Entire α is specified to left tail of the critical region

Points Left



Large Samples Test's

Under large sample tests, we will see four important tests. Test the significance of

1. Single mean
2. Difference of means
3. Single proportion
4. Difference of proportions

1. Testing a Claim About a Mean

Key Concept

Suppose that we wish to test the hypotheses

$$H_0 : \mu = \mu_0$$

Requirements:

1) The sample is a simple random sample.

2) Either or both of these conditions is satisfied: The population is normally distributed or *n* > 30.

Under null hypothesis

$$(i)\, H_1 : \mu \neq \mu_0 \text{ or}$$

$$(ii)\, H_1 : \mu > \mu_0 \text{ or}$$

$$(iii)\, H_1 : \mu < \mu_0$$

Test Statistic for single mean

Condition	Test Statistic
When σ is known	$Z_{cal} = \dfrac{\bar{X} - \mu}{\sigma / \sqrt{n}}$
When σ is unknown s is s.d of sample mean	$Z_{cal} = \dfrac{\bar{X} - \mu}{s / \sqrt{n}}$

Problems on Single mean

1. The life span of electric bulbs manufactured by a particular company follows a normal distribution with a standard deviation of 120 hours and its life span is guaranteed under warranty for a minimum of 800 hours. At random, a sample of 50 bulbs from a lot is selected and it is revealed that the life span is 750 hours. With a significance level of 0.01, should the lot be rejected by not honoring the warranty?

- Solution:
- σ=120 hrs
- life span is guaranteed under warranty for a minimum of 800 hours.
- μ<800 hrs
- Sample of Size n=50 $\bar{x} = 750\,hrs$
  - α= 1%
  - σ=120 hrs

1. The life span of electric light bulbs … With a significance level of 0.01, should the lot be rejected by not honoring the warranty?

- 1. **State H_0** $H_0: \mu = 800$ hrs
- 2. **State H_1** $H_1: \mu < 800$ hrs
- 3. **Given that α** α = 0.01
- 4 **Set Up Critical Value(s)** $Z_{tab} = -2.33$
- 5 **Compute Test Statistic :**

$$Z_{cal} = \frac{750 - 800}{120 / \sqrt{50}}$$

$Z_{cal} = -2.94$

- 6 **Make Statistical Decision:** $|Z_{cal}| > |Z_{tab}|$

Test statics falls in rejection region. So, *Reject Null H_0*

- **7. Express Decision:** should the lot be rejected by not honoring the warranty? No

2. Test of significance for difference of means

Requirements:

1) **The samples are simple and independent random samples.**

2) **Either or both of these conditions is satisfied: The populations are normally distributed or large samples**

Under null hypothesis $H_0 : \mu_1 = \mu_2$

$(i)\, H_1 : \mu_1 \neq \mu_2$ or

$(ii)\, H_1 : \mu_1 > \mu_2$ or

$(iii)\, H_1 : \mu_1 < \mu_2$

Test Statistic for difference of means

Condition	Statistic
When σ_1 and σ_2 are known	$Z = \dfrac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}$
When σ_1 and σ_2 are unknown	$Z = \dfrac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$

3. Testing a Claim About a Proportion

Requirements:

1) The sample observations are a simple random sample.

2) The conditions for a binomial distribution are satisfied .

3) The conditions $np \geq 5$ and $nq \geq 5$ are satisfied, so the binomial distribution of sample proportions can be approximated by a normal distribution with $\mu = np$ and $\sigma = \sqrt{npq}$.

Test Statistic for single proportion

Under null hypothesis $H_0 : P = P_0$

$(i)\, H_1 : P \neq P_0$ or

$(ii)\, H_1 : P > P_0$ or

$(iii)\, H_1 : P < P_0$

$$Z = \frac{p - P_0}{\sqrt{\dfrac{P_0 Q_0}{n}}}$$

4. Test of significance for difference of proportions

Requirements:

1. Suppose we want to compare two distinct populations with respect to the prevalence of a certain attribute, say *A*, *among their members.*
2. Let x_1, x_2 *be the* number of persons possessing the given attribute *A in random samples of sizes* n_1 and n_2 *from the two populations respectively*.
3. If P_1 and P_2 are the populations proportions, then $E(p_1)=P_1$ and $E(p_2)=P_2$.
4. **Under null hypothesis**

$$H_0 : P_1 = P_2$$

$$(i)\, H_1 : P_1 \neq P_2 \text{ or}$$

$$(ii)\, H_1 : P_1 > P_2 \text{ or}$$

$$(iii)\, H_1 : P_1 < P_2$$

Test Statistic for difference of means

Condition	Statistic
Samples are taken from different population	$Z = \frac{(p_1 - p_2) - (P_1 - P_2)}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}$
Samples are taken from same population	$Z = \frac{(p_1 - p_2)}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$
Samples are taken from same population. But, population proportion ***P*** is unknown	$Z = \frac{(p_1 - p_2)}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$ $\quad P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$